

Gauge Historical Daily Synthetic Generation Model (MKM-GHD-001)

Wolfram Oracle Benchmark: the Series' Statistics Computed Exactly

MKM Research Labs

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1 Executive Summary

This document benchmarks the Gauge Historical Daily Synthetic Generation Model (MKM-GHD-001), which generates a 50-year daily water-level history for each gauge from its flood thresholds, tidal regime and recorded count of severe exceedances. Those histories feed the gauge statistics and, through their monthly means, the stress pipeline. It asks whether the code generates what its model document (`gaugehd_synthetic.pdf`, v1.0) says, and whether what the document says about the result holds.

The code generates its documented process. The model is random, so it is judged in distribution. For six gauges (each tidal regime, with no recorded severe exceedance and with eight), the expected number of days at or above each flood threshold, the mean level, and the January and July departures from it are computed exactly from the document's equations (1)–(14). Each is then compared with the mean over 200 generated series of 18,251 days. Every statistic agrees within sampling error, worst $|z| = 2.03$ (Tables 1 and 2). Wolfram's exact integrals agree with the reference's double-precision closed forms to 4.9×10^{-10}.

But several of the document's own statements about the output do not hold (Table 3):

1. **Severe frequency is understated tenfold (§4.5).** `FrequencyExceedLevel3` counts severe exceedances over five years (Table 2), but Eq. (9) spreads that count over the whole 50-year run. A gauge with eight severe exceedances in five years is given eight in fifty. This is the model's main control on the upper tail (§6.4).

2. **Regeneration is not reproducible (§5.4, §7.3).** The seed is Python's `hash()` of the gauge id, which is salted per process: the same id seeds 4181464545 in one process and 3341949591 in another. The series also ends on the day it is run, so even a fixed seed gives a different calendar each day.

3. **The flood-frequency diagnostic cannot pass (§7.1(3)).** It expects about $6\max(n_3, 1)$ days above FloodAlert. The exact expectation is 16.3 and 129.7 for $n_3 = 0$ and 8, because a flood lifts the AR(1) state and the level stays above the alert for 2.7 days on average (non-tidal). For the same reason, §8.4's single-day spikes are not what the model produces.

4. **The lag-1 diagnostic cannot pass (§7.1(1)).** The AR(1) noise has lag-1 autocorrelation 0.85, but the series is dominated by its seasonal cycle. Its lag-1 autocorrelation is 0.981 without floods, and 0.967 as generated.

5. **The spring–neap cycle restarts each year (§3.5).** Eq. (6) takes the phase from the day of the year, so it jumps 9.8 days each 1 January.

The seasonal means (§5.6), the lag-7 autocorrelations (§6.1) and the mean levels (§7.2) are as stated.

Evidence. 7 checks, outcome: all pass (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test, at commit `7bbb63078a59`.
- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-10-28 by `python -m tools.wolfram_oracle ghd001`.

2 Model Purpose and Scope

2.1 What is benchmarked

`generate_synthetic_timeseries` in `src/models/statistics/synthetic.py`, imported as production imports it and called with a gauge record as `src/port/src/gauge/gaugehd/synthetic.py` builds one, on the document's default thresholds (3, 4 and 5 m) and a historical high of 5.5 m.

2.2 What is not

The historical-high injection (§3.8) and the 0.1 m floor (§3.9), which these gauges never reach. Also out of scope: the tidal classification by longitude (§3.2), which is a lookup; the statistics module, which is the Timeseries Statistics Model's subject; and whether synthetic histories are a sound stand-in for observed ones (§7.4). This is a benchmark of the implementation (validation question Q8), not independent validation (Q9).

3 Mathematical Framework

Following the document, a day's level is $W_t = \mu_0 + S_t + T_t + \varepsilon_t$ unless a flood is injected, with $\mu_0 = h_{\text{alert}} U(0.3, 0.5)$, $S_t = A_s \cos(2\pi(d_t - 15)/365)$, $T_t = A_\tau M_t V_t$ with $V_t \sim U(-1, 1)$, and ε_t an AR(1) with coefficient ϕ and marginal deviation σ_w. A severe flood occurs with daily probability $p = \max(n_3, 1)/N$ and a minor one with probability $5p$. Each sets the level uniformly in its band, and resets ε to $R = W_t - \mu_0$.

The exact expectation. If the last flood was k days earlier, then $\varepsilon_t = \phi^k R + \sigma_w \sqrt{1 - \phi^{2k}}\, Z$. So, given μ_0, the day's level is normal about $\mu_0 + S_t + \phi^k R + T_t$, with R and V_t uniform. The chance that it reaches a threshold h is

$$P(W_t \geq h \mid k) = \mathbb{E}\Big[Q\Big(\tfrac{h - \mu_0 - S_t - \phi^k R - A_\tau M_t V_t}{\sigma_w \sqrt{1 - \phi^{2k}}}\Big)\Big], \tag{1}$$

a normal tail Q averaged over two uniforms. Integrating twice gives a closed form in $G(z) = \int_z^\infty Q$ and $H(z) = \int_z^\infty G$. The age k is geometric, with the flood probabilities as its parameter, and the levels are rounded to the millimetre as reported. Summing over the run's calendar gives each expected count and the expected mean, January and July levels. The base level μ_0 is then integrated by 24-point Gauss–Legendre.

4 Benchmark Method

Each gauge is generated 200 times, with seeds 1 to 200 passed explicitly, so that the run does not depend on the per-process hash. Each statistic's mean over the series is compared with its exact expectation in standard errors, and a check passes at $|z| \leq 4$. The six gauges share their seeds, and so their base-level draws, so their errors are correlated. That is why the mean level sits about one standard error low on all six at once. The reference is evaluated in double precision. Wolfram derives the double integral above symbolically with `Integrate`, from each input's exact value, and evaluates

it to 30 digits at 18 points (three regimes, six days and ages), down to tails of 10^{-53}. There it is compared with the reference's closed form.

The claims are judged from the exact expectations, not from the samples, except for reproducibility, which hashes one gauge id in two fresh Python processes with different `PYTHONHASHSEED`s.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Limit	Outcome
Days at or above FloodAlert: model against the exact expectation, worst $\|z\|$	1.39	4	pass
Days at or above FloodWarning: model against the exact expectation, worst $\|z\|$	2.03	4	pass
Days at or above SevereFloodWarning: model against the exact expectation, worst $\|z\|$	1.92	4	pass
Mean level: model against the exact expectation, worst $\|z\|$	1.02	4	pass
January mean less the annual mean: model against the exact expectation, worst $\|z\|$	1.37	4	pass
July mean less the annual mean: model against the exact expectation, worst $\|z\|$	1.40	4	pass
Wolfram's exact integrals agree with the closed forms	4.9×10^{-10}	1.0×10^{-9}	pass

5.1 The gauges

Table 2: Exact expectation / model mean over 200 series, per 50-year series

Gauge	Floods	Days $\geq$ alert	$\geq$ warning	$\geq$ severe	Mean level (m)
Non-tidal/n3=0	6.0	16.26 / 15.82	2.62 / 2.42	1.02 / 0.96	1.206 / 1.193
Non-tidal/n3=8	48.0	129.67 / 129.42	20.93 / 20.68	8.19 / 8.14	1.245 / 1.232
Partially tidal/n3=0	6.0	10.52 / 10.12	1.70 / 1.48	1.00 / 0.88	1.204 / 1.191
Partially tidal/n3=8	48.0	84.04 / 84.02	13.55 / 13.31	8.00 / 7.88	1.227 / 1.215
Tidal/n3=0	6.0	7.64 / 7.32	1.06 / 0.96	1.00 / 0.88	1.202 / 1.190
Tidal/n3=8	48.0	61.00 / 61.09	8.49 / 8.24	8.00 / 7.88	1.217 / 1.204

5.2 The model document's claims

Table 3: The model document's claims, tested

Claim	Stated	Benchmark finds	Outcome
§7.1(1)	lag-1 autocorrelation ≈ 0.85 (non-tidal)	0.981 without floods; the model gives 0.967 and 0.916: the seasonal cycle dominates	**not met**
§7.1(3)	days above FloodAlert $\approx 6\max(n_3, 1)$	16.3 and 129.7 for $n_3 = 0, 8$, against 6 and 48	**not met**
§8.4	single-day flood spikes	days above FloodAlert per flood: 2.70, 1.75, 1.27 (non-tidal, partial, tidal)	**not met**
§5.6	January, July means $\approx \pm A_s$	0.985–0.988 of A_s	met
§4.5	reproduces the observed severe frequency	n_3 counts 5 years but is spread over 50: 8 observed per 5 years become 0.8	**not met**
§5.4, §7.3	byte-identical regeneration	one gauge id seeds 4181464545 in one process and 3341949591 in another; the series also ends at the run date	**not met**
§6.1	lag-7 autocorrelations 0.32, 0.028	0.3206, 0.0280	met
§7.2	mean levels below FloodAlert	at most 1.245 m	met
§3.5	a 14.8-day spring–neap cycle	its phase is the day of the year, so it jumps 9.8 days each 1 January	finding

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The code is shown to generate its documented process. Whether that process resembles a Thames gauge is §7.4's backtest, which this does not perform.

2. **Severe frequency.** Eq. (9) should divide n_3 by the five years it counts, not by Y; or §4.5 and Table 2 should say what n_3 is meant to be. This raises every gauge's flood rate tenfold, and with it the upper tail of every generated history, so it is for the model owner and the MRC.

3. **Reproducibility.** A stable seed (for example from `hashlib`), and a fixed end date taken from the portfolio rather than the clock.

4. **Diagnostics.** §7.1(1) should test the AR(1) residual, not the series. §7.1(3) and §8.4 should allow for the days a flood takes to decay. §3.5 should either take the phase from a running day count or say it restarts each year.

7 Document History

Date	Version	Description	Author
28-Oct-2026	0.1	First issue	David K Kelly